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#include <stdio.h>

#include <omp.h>

// Function to check if a number is prime
int is_prime(int n)
{
    if (n < 2) return 0;

    for (int i = 2; i * i <= n; i++)
    {
        if (n % i == 0) return 0;
    }

    return 1;
}

int main()
{
    long n = 500000; // Adjusted default range for quick execution (you can increase this)

    double start, end;

    double time_serial, time_parallel;

    printf("\nEnter the upper limit (n): ");
    if (scanf("%ld", &n) != 1 || n <= 0) {
        printf("Invalid input!\n");
        return 1;
    }

    printf("\nChecking prime numbers in range 1 to %ld\n", n);
    printf("-----\n");

    // ----- SERIAL EXECUTION -----

    start = omp_get_wtime();

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for (int i = 1; i <= n; i++)
{
    is_prime(i);
}

end = omp_get_wtime();

time_serial = end - start;

printf("Time to compute prime numbers (Serial) : %.6f seconds\n", time_serial);


// ----- PARALLEL EXECUTION -----

start = omp_get_wtime();

#pragma omp parallel for schedule(dynamic)
for (int i = 1; i <= n; i++)
{
    is_prime(i);
}

end = omp_get_wtime();

time_parallel = end - start;

printf("Time to compute prime numbers (Parallel) : %.6f seconds\n", time_parallel);


printf("-----\n");

printf("Speedup Ratio : %.2fx\n", time_serial / time_parallel);

printf("-----\n");


return 0;
}

```